

DYNAMIC DIMENSIONAL REDUCTION IN THE ABELIAN SANDPILE

AHMED BOU-RABEE

ABSTRACT. We prove the dimensional reduction conjecture of Fey, Levine, and Peres (2010) on the hypercube. The proof shows that dimensional reduction, symmetry, and regularity of the Abelian sandpile persist during the parallel toppling process. This stronger result verifies empirical observations first documented by Liu, Kaplan, and Gray (1990).

1. INTRODUCTION

1.1. Overview. Let $\mathcal{C}_N^{(d)}$ be a hypercube of side length N centered at the origin in the d -dimensional integer lattice $\mathbb{Z}^d$. A *sandpile* is a function $s : \mathcal{C}_N^{(d)} \rightarrow \mathbb{Z}$. Start with $2d$ chips in the hypercube, $s = s_0^{(d)} \equiv 2d$, then iterate the following rule: every site with at least $2d$ chips on it becomes *unstable* and *topples* in *parallel*, simultaneously giving one chip to each of its $2d$ neighbors. If a boundary site topples, it loses chips over the edge. Eventually every site is stable and the process stops, yielding a sequence of sandpiles over time $\{s_t^{(d)}\}_{t \geq 0}$ [BTW87, Dha90, HLM⁺08, LP10, LP17, Jár18, Kli18].

Simulations suggest an exact relationship between these sandpiles across size N , time t , and dimension d — see Figures 1 and 2. Specifically, it appears that smaller sandpiles are embedded in larger sandpiles of the same dimension at certain times. Moreover, central cross sections of d -dimensional sandpiles coincide almost exactly with $(d - 1)$ -dimensional sandpiles for all time.

In this article, we provide a rigorous proof of these observations, *self-similarity* and *dimensional reduction*, via a simultaneous induction on all parameters: size, time, and dimension. Along the way, we develop new techniques for analyzing the parallel toppling process which may be of independent interest.

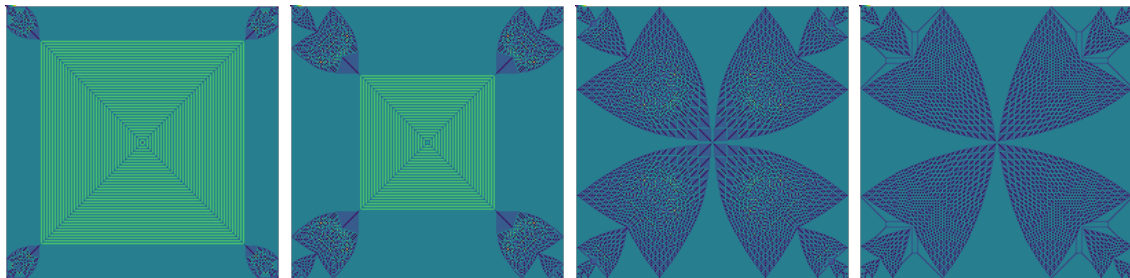


FIGURE 1. $s_t^{(2)}$ for $t = (25)^2, (50)^2, (100)^2, \infty$, where $N = 200$ and $s_0 \equiv 4$. Sites with $0, \dots, 7$ chips are represented by different colors.

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FIGURE 2. $s_t^{(3)}(\cdot, \cdot, \text{offset})$ for $\text{offset} = 0, 20, 40, 60$, where $t = (50)^2$, $N = 200$, and $s_0 \equiv 6$. Sites with $0, \dots, 11$ chips are represented by different colors and ‘offset’ is distance to the center of $\mathcal{C}_N^{(3)}$.

Dimensional reduction in sandpiles has been known experimentally since at least the work of Liu, Kaplan, and Gray in 1990 [LKG90]. In 2010, Fey, Levine, and Peres formulated an approximate dimensional reduction conjecture in the context of the single-source sandpile [FLP10] — this was later highlighted in a survey by Levine and Propp [LP10]. Soon after, Pegden and Smart, in a breakthrough, used the theory of viscosity solutions to show that the large-scale patterns which appear in sandpiles can be described by a limiting *sandpile PDE* [PS13]. This led to a series of works, joint with Levine, which provide a remarkable understanding of sandpiles on $\mathbb{Z}^2$ [LPS17, LPS16, PS20].

The structure of the sandpile PDE reveals that on torus-like domains one can construct limit sandpiles in $\mathbb{R}^d$ from those in $\mathbb{R}^{d-1}$. This is also possible for certain random sandpiles [BR21]. However, it has remained completely open to prove (or disprove) dimensional reduction for any natural example, on either the lattice or the continuum, till now.

Our proof in this paper leverages discrete techniques to establish dimensional reduction on the hypercube when $s_0 \equiv 2d$. The main insight is recognizing that the parallel toppling process together with strong induction can be used to control the sandpile as it stabilizes. We identify a delicate interplay between the initial condition and the symmetry of the hypercube which forces the ‘flow’ of parallel toppling to preserve dimensional reduction. In fact, we do not know how to prove dimensional reduction for only the terminal sandpile. Our proof involves no technology from viscosity solutions nor knowledge of the sandpile PDE.

A key step in the proof is the application of certain discrete derivative bounds — Corollary 2.2 and Lemma 2.5 below. The strength of these bounds deteriorates as the initial condition grows and it is only when $s_0 \equiv 2d$ that they are strong enough to ensure dimensional reduction.

This dependence on the initial condition provides some explanation for what occurs in the single-source version. Fey-Levine-Peres conjectured that when the initial condition is $(2d - 2) + n\delta_0$ dimensional reduction occurs after carving out a region near the origin. This is likely because our derivative bound (which, in the single-source version, has a flipped dependence on the constant background) does not hold near the origin, but only holds away from it. Proving this remains open, but we believe the methods in this paper extend to this and other initial data.

In fact, the result in this paper appears to be a special case of a more general one which we cannot yet prove: dimensional reduction occurs for any uniform initial condition in high enough dimensions. Specifically, for integer $k \geq 0$, when $s_0 \equiv 2d + k$, above a critical dimension, $d > d_0 := k + 1$, exact dimensional reduction, modulo $(d_0 - 1)$ -dimensional

defects, appears to persist throughout the parallel toppling process. We show that when $d = d_0$, dimensional reduction fails to occur, providing one explanation for why $(d_0 - 1)$ -dimensional defects appear — see Table 1.

1.2. Main Result. Our proof begins with the *odometer* v_t , which encodes the number of topples at each site over time. Let $v_0 : \mathcal{C}_N^{(d)} \rightarrow \mathbb{Z}^+$ be the initial odometer $v_0 \equiv 0$; then, recursively,

$$\begin{aligned} (1) \quad & v_{t+1} = v_t + 1\{s_t \geq 2d\}, \\ (2) \quad & s_{t+1} = s_t + \Delta(v_{t+1} - v_t), \end{aligned}$$

where Δ is the graph Laplacian on $\mathcal{C}_N^{(d)}$ with dissipating boundary conditions. Dependence on d and N is indicated by writing $v_t^{(d,N)}$ and $s_t^{(d,N)}$.

We prove dimensional reduction and self-similarity of sandpiles via an analysis of the parallel toppling odometers. It will be more convenient to state the result after making a symmetry reduction. Let $\text{Aut}_{\mathcal{C}_d}$ denote the group of symmetries of the d -dimensional hypercube and let it act on $\mathbb{Z}^d$ by matrix-vector multiplication. The definitions imply that parallel toppling preserves the symmetries of the hypercube: $v_t^{(d,N)}(\mathbf{x}) = v_t^{(d,N)}(\sigma\mathbf{x})$ for all $t \geq 1$, $\mathbf{x} \in \mathcal{C}_N^{(d)}$, and $\sigma \in \text{Aut}_{\mathcal{C}_d}$ (we give a simple proof in Section 2.4 below). Hence the odometer and sandpile are fully determined by their restrictions to a fundamental domain of the hypercube — we choose a particular one in the statement of the theorem.

Theorem 1.1. *Let $N \geq 1$, $d \geq 2$, and $M = \lceil N/2 \rceil$. Denote the fundamental domain of $\mathcal{C}_N^{(d)}$ consisting of sorted coordinates in decreasing order by $\mathcal{S}_M^{(d)} := \{(x_1, \dots, x_d) : M \geq x_1 \geq \dots \geq x_d \geq 1\}$.*

(1) *Dimensional reduction: for all $t \geq 1$ and $\mathbf{x}_{d-1} \in \mathcal{S}_M^{(d-1)}$*

$$v_t^{(d,N)}(\mathbf{x}_{d-1}, 1) = v_t^{(d-1,N)}(\mathbf{x}_{d-1})$$

and for $\mathbf{x}_{d-1} \geq 2$

$$v_\infty^{(d,N)}(\mathbf{x}_{d-1}, 2) = v_\infty^{(d-1,N)}(\mathbf{x}_{d-1}).$$

(2) *Self-similarity: for all $j < M$, $t \leq \tau_j$, and $\mathbf{x} \in \mathcal{S}_M^{(d)}$ with $\mathbf{x} > M - j$*

$$v_t^{(d,N)}(\mathbf{x}) = v_t^{(d,2j)}(\mathbf{x} - (M - j)),$$

where

$$\tau_j := \min\{t \geq 1 : v_t^{(d,2j)}(\mathbf{x}) \geq j \text{ for } \mathbf{x} \in \partial\mathcal{C}_{2j}^{(d)}\},$$

and $\partial\mathcal{C}_i^{(d)} := \{\mathbf{x} \in \mathcal{C}_i^{(d)} : \text{there is } \mathbf{y} \notin \mathcal{C}_i^{(d)} \text{ with } |\mathbf{y} - \mathbf{x}| = 1\}$ denotes the inner boundary of the cube.

Both of these results immediately translate to the sandpile.

Corollary 1.1. *The results in Theorem 1.1, using the same notation, extend to the sequence of sandpiles.*

(1) *Dimensional reduction: for all $\mathbf{x}_{d-1} \in \mathcal{S}_M^{(d-1)}$ with $\mathbf{x}_{d-1} \geq 2$*

$$s_\infty^{(d,N)}(\mathbf{x}_{d-1}, 1) = s_\infty^{(d-1,N)}(\mathbf{x}_{d-1}) + 2.$$

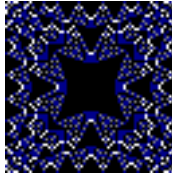
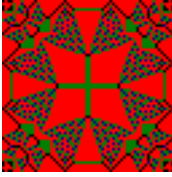
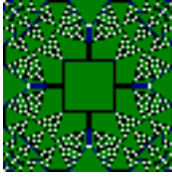
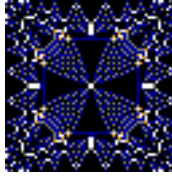
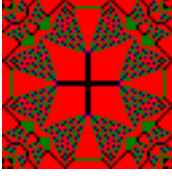
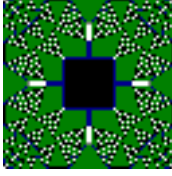
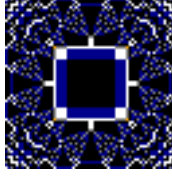
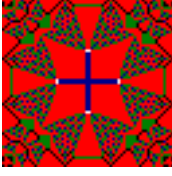
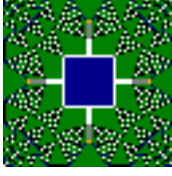
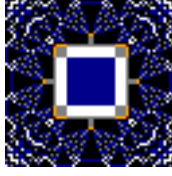
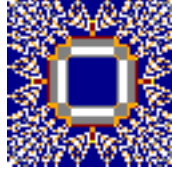
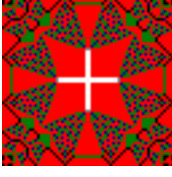
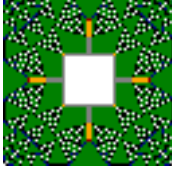
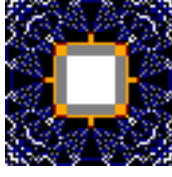
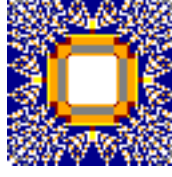
k	0	1	2	3	4
$s_\infty^{(2)}(\cdot, \cdot)$					
$s_\infty^{(3)}(\cdot, \cdot, 1)$					
$s_\infty^{(4)}(\cdot, \cdot, 1, 1)$					
$s_\infty^{(5)}(\cdot, \cdot, 1, 1, 1)$					
$s_\infty^{(6)}(\cdot, \cdot, 1, 1, 1, 1)$					

TABLE 1. Center slices of terminal sandpile configurations for $s_0 \equiv 2d + k$ and $N = 64$. Site colors are normalized by column so that in dimension d a site with z chips has the same color as a site with $(z - 2)$ chips in dimension $(d - 1)$.

(2) *Self-similarity:* for all $j < M$, $t \leq \tau_j$, and $\mathbf{x} \in \mathcal{S}_M^{(d)}$ with $\mathbf{x} > M - j + 1$

$$s_t^{(d,N)}(\mathbf{x}) = s_t^{(d,2j)}(\mathbf{x} - (M - j)).$$

The first part of Theorem 1.1 states that the parallel toppling odometer restricted to a center slice of a d -dimensional hypercube coincides with the $(d - 1)$ -dimensional parallel toppling odometer for all time. Dimensional reduction holds for the final odometer off the center, implying dimensional reduction on the center for the final, stable sandpile.

The second part of the theorem relates the parallel toppling odometers for different sized cubes: the odometer for the size N cube contains the size $2j$ cube odometer up until the first time a boundary site exceeds j topples. These automatically imply, after shrinking the domains by one, the same for the sequence of sandpiles.

As mentioned previously, we expect Theorem 1.1 to be a special case of a more general result. In Section 3 we show that dimensional reduction does not occur when $s_0 \equiv 2d + (d - 1)$

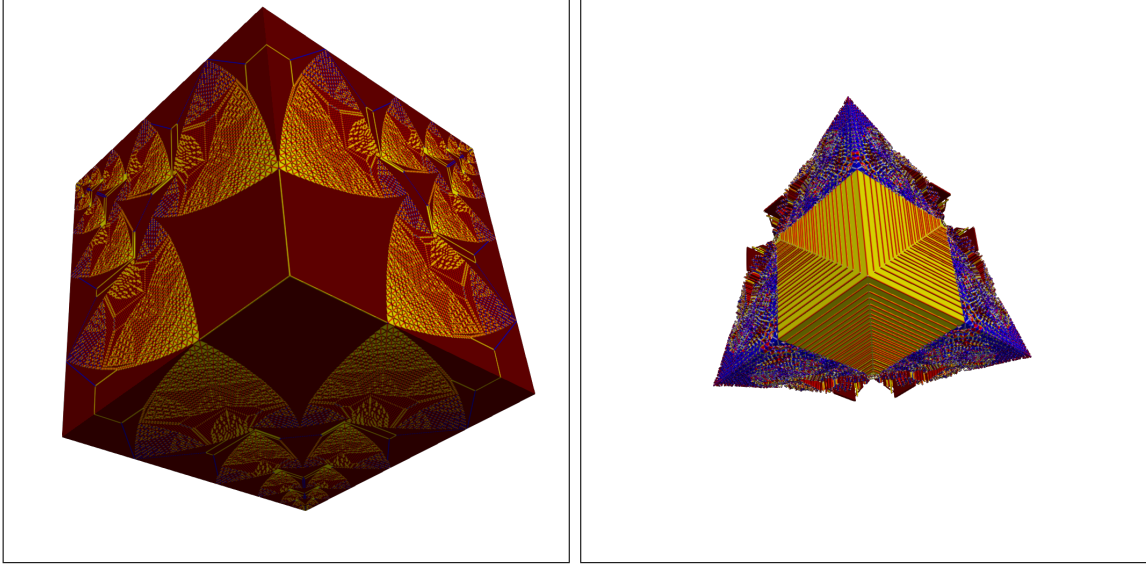


FIGURE 3. On the left, a corner cube of $s_{\infty}^{(N,3)}$ and on the right the same corner with all cross sections which match $s_{\infty}^{(N,2)}$ removed. The initial condition is $s_0 \equiv 7$ and $N = 2^{11}$. Sites with $0, \dots, 5$ chips are represented by different colors.

in all dimensions $d \geq 2$ when $N = 2$. We also provide an explicit description of the parallel toppling odometer when $s_0 \equiv 2d + (d - 1)$ and $N = 4$ for all $d \geq 1$. This explicit form suggests that the proof template in this paper may help with the following.

Problem 1.2. *Show that Theorem 1.1 holds when $s_0 \equiv 2d + k$ for all $k \geq 0$ and $d > d_0 := (k + 1)$.*

We expect an even stronger result to be true, although it is likely the proof will require techniques beyond those presented here. In simulations, exact dimensional reduction appears to occur away from the central slice. For example, when $s_0 \equiv 4$, $d = 2$, and N is large, the center of the sandpile contains large curved triangles of 3s. In fact, for every dimension and size we could simulate, whenever dimensional reduction occurs along the central cross sections of the hypercube, it extends; see Figure 3 for an example in three dimensions.

Problem 1.3. *Extend dimensional reduction on the hypercube to a domain of codimension zero.*

The following is closely related.

Problem 1.4. *Show that the odometer for any bounded initial sandpile on the hypercube has bounded second differences. For instance, show for all $t \geq 1$ and $i = 1, 2$ that*

$$-3 \leq -2v_t(\mathbf{x}) + v_t(\mathbf{x} + e_i) + v_t(\mathbf{x} - e_i) \leq 2$$

when $d = 2$ and $s_0 \equiv 4$.

Numerical evidence indicates the hypercube is a necessary hypothesis in Problem 1.4. In fact, in most other domains, including the discrete circle, the odometer does not appear to have bounded second differences. On the hypercube, our proof of dimensional reduction

shows that the odometer has bounded second differences along the central cross sections; however, a method to propagate those bounds to the interior remains out of reach.

1.3. Outline of the proof. The proof of Theorem 1.1 is a careful induction on hypercube dimension, side length, and time. Some parts of the argument can be simplified but we present it in this fashion to suggest a template for proving dimensional reduction with more general initial data. At a high level, we show that if the parallel toppling process for a d_0 -dimensional sandpile is sufficiently regular, then dimensional reduction is guaranteed in all dimensions $d > d_0$. We prove this regularity when $d_0 = 1$; the case $d_0 > 1$ remains open.

Our main technical tool is a technique introduced by Babai and Gorodezky to prove discrete quasiconcavity of the single-source sandpile odometer in $\mathbb{Z}^2$ [BG07]. By an iteration of their technique, we gain symmetry of the odometer, a derivative comparison result, and a parabolic least action principle. These results, which appear in Section 2, extend beyond the hypercube and so may be of independent interest.

In Section 3 we explicitly determine $v_t^{(d,N)}$ when $N = 4$ in all dimensions $d \geq 1$ when the initial sandpile is $s_0 \equiv 2d + d - 1$. This is done by mapping the hypercube to a line via a radial decomposition. The explicit form of v_t provides both a base case for our proof and progress towards Problem 1.2. We also show that when $N = 2$ dimensional reduction does not occur at the critical dimension d_0 .

The explicit solution when $N = 4$ establishes the base case for an odometer regularity result which is then proved in Section 4. Finally, in Section 5, we use the established regularity of the odometer in dimension $(d - 1)$ to prove dimensional reduction in dimension d .

An efficient algorithm for computing high-dimensional sandpiles. In Section 2.4 we show that $v_t^{(d,N)}$ can be computed via the parallel toppling procedure restricted to the simplex. In fact, the argument shows that any sandpile with a symmetric initial condition on $\mathbb{Z}^d$, including the single-source sandpile, retains symmetry throughout the parallel toppling process and can be computed in this way.

For d large, computing sandpiles on the simplex improves space complexity by a factor of d^d . Moreover, the reduction in size also leads to a faster algorithm when using parallelization. We wrote a program in Julia [BEKS17, BFDS18] for computing arbitrary dimensional sandpiles which implements these improvements. The program, which may be freely used and modified, is included in the arXiv post.

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2. PRELIMINARIES

2.1. Notation and conventions. When we need to distinguish between vectors and scalars, we reserve $\mathbf{x}$ for vectors and x for scalars. The i -th element of $\mathbf{x}$ is x_i and $\mathbf{x}_i = (x_1, \dots, x_i)$. We refer to coordinate basis vectors of $\mathbb{Z}^d$ as e_i and the ones vector of length i by $\mathbf{1}_i = (1, \dots, 1)$. Equalities, inequalities, addition, and multiplication between vectors and scalars are to be

understood pointwise. The notation $|x|$ refers to the absolute value and $|\mathbf{x}| = \sum_{i=1}^d |x_i|$ is the ℓ_1 norm.

We embed $\mathcal{C}_N^{(d)}$ into $\mathbb{Z}^d$ in two different ways depending on whether N is even or odd. If $N = 2k + 1$, $\mathcal{C}_N^{(d)} = \{\mathbf{x} + 1 : |x_i| \leq k\}$ otherwise $\mathcal{C}_N^{(d)} = \{\mathbf{x} - k : 1 \leq x_i \leq 2k\}$. The graph Laplacian on $\mathcal{C}_N^{(d)}$ operates on functions $f : \mathcal{C}_N^{(d)} \rightarrow \mathbb{R}$ as

$$(3) \quad \Delta^{(d)} f(x) = -2df(x) + \sum_{y \sim x} f(y),$$

where we pad $f(x) := 0$ for $x \notin \mathcal{C}_N^{(d)}$ and the sum $y \sim x$ is over the $2d$ nearest neighbors of x , $|y - x| = 1$. When the hypercube size or dimension is not used, we omit distinguishing sub/superscripts.

2.2. Babai-Gorodezky technique. In this subsection and the next, let $s_0 \leq 2(2d) - 1$ be an arbitrary initial sandpile on $\mathcal{C}$ and v_t its odometer. A straightforward induction argument and the definition of the graph Laplacian yields the following lemma.

Lemma 2.1. *For each $x \in \mathcal{C}$ and all $t \geq 0$,*

$$(4) \quad v_{t+1}(x) = \lfloor \frac{s_0(x) + \sum_{y \sim x} v_t(y)}{2d} \rfloor.$$

Babai and Gorodezky used this simple lemma to prove a nontrivial discrete quasiconcavity property of the single-source sandpile in $\mathbb{Z}^d$ [BG07]. A more general version of their argument appears below in Lemma 2.4. Roughly, their technique recognizes that if a property of the odometer holds at $t = 1$, is consistent across the symmetry axes, and can be verified on the boundaries of the domain, it must hold for all $t \geq 1$.

Lemma 2.1 is used many other times throughout this paper; notably we use it to prove a parabolic least action principle and symmetry of the odometer on the hypercube.

2.3. Parabolic least action principle. The least action principle [FLP10] shows that v_∞ is minimal among all $w : \mathcal{C} \rightarrow \mathbb{Z}^+$ which stabilize s_0 : $\Delta w + s_0 \leq 2d - 1$. We upgrade this to a parabolic least action principle by observing the parallel toppling procedure as a directed sandpile on $\mathcal{G} = \mathbb{Z}^+ \times \mathcal{C}$. The initial sandpile and odometer over time are stacked, $s(t, x) := s_0(x)$ and $v(t, x) := v_t(x)$ for all $t \geq 0$ and $x \in \mathcal{C}$. The graph Laplacian operates on functions $f : \mathcal{G} \rightarrow \mathbb{R}$ as

$$(5) \quad \Delta^{\mathcal{G}} f(t, x) = -2df(t, x) + \sum_{y \sim x} f(t - 1, y),$$

for $t \geq 1$ and $x \in \mathcal{C}$, where the sum $y \sim x$ is over the nearest neighbors of x in $\mathcal{C}$.

Lemma 2.2 (Parabolic least action principle).

$$(6) \quad v(t, x) = \min\{u : \mathcal{G} \rightarrow \mathbb{Z}^+ : \Delta^{\mathcal{G}} u(t, x) + s(t, x) \leq 2d - 1 \text{ for all } x \in \mathcal{C} \text{ and } t \geq 1\}$$

Proof. Let $w(t, x)$ denote the right-hand side of (6). We show using Lemma 2.1 and induction that $v(t, x) = w(t, x)$. Indeed, by the directed structure of $\mathcal{G}$, it suffices to show this equality one time slice at a time. Equality holds at $t = 0$ as $v(0, x) = w(0, x) = 0$. Assume that

$v(t', \cdot) = w(t', \cdot)$ for $t' \leq t$ and let $x \in \mathcal{C}$ be given. The monotonicity of the graph Laplacian implies $\Delta^{\mathcal{G}}w + s \leq 2d - 1$, hence,

$$\Delta^{\mathcal{G}}w(t+1, x) + s(t+1, x) = -2dw(t+1, x) + s_0(x) + \sum_{y \sim x} w(t, y) < 2d,$$

and a rearrangement yields,

$$w(t+1, x) \geq \lfloor \frac{s_0(x) + \sum_{y \sim x} w(t, y)}{2d} \rfloor.$$

By Lemma 2.1 and the inductive hypothesis, the right-hand side of the above is exactly $v(t+1, x)$. Similarly, for the other direction,

$$\begin{aligned} \Delta^{\mathcal{G}}v(t+1, x) + s(t+1, x) &= 2d \left(\frac{s_0(x) + \sum_{y \sim x} w(t, y)}{2d} - \lfloor \frac{s_0(x) + \sum_{y \sim x} w(t, y)}{2d} \rfloor \right) \\ &< 2d, \end{aligned}$$

which concludes the proof by minimality of w . $\square$

Our usage of the parabolic least action principle in the main argument is minimal and can be avoided. And, in some sense, it is a restatement of Lemma 2.1. We included it as it may be of independent interest.

2.4. Symmetry and fundamental domains. In this section we observe that sandpile dynamics on $\mathcal{C}_N^{(d)}$ preserve the symmetry structure of the d -dimensional hypercube. This is then used to reduce to the sandpile on a *fundamental domain* of the hypercube with reflecting boundary conditions. The main contribution of this subsection is a coordinate-wise description of this domain along with an explicit formula for the reflecting graph Laplacian.

We briefly provide a presentation of the group of automorphisms of the hypercube and its action on $\mathbb{Z}^d$; for more details see, for example, [GR13]. Let $\text{Aut}_{\mathcal{C}_d}$ be the group of $(d \times d)$ matrices with exactly one ± 1 in each row and in each column and 0s elsewhere. Let $\sigma \in \text{Aut}_{\mathcal{C}_d}$ act on $x \in \mathbb{Z}^d$ by matrix-vector multiplication followed by a translation and let it act on $f : \mathbb{Z}^d \rightarrow \mathbb{R}$ by $\sigma f(x) := f(\sigma x)$. The translation is chosen to preserve $\mathcal{C}_N^{(d)}$ when N is even or odd in our choice of coordinates.

Each σ is an isometry and hence preserves nearest neighbors and $\mathcal{C}_N^{(d)}$. That is, if $y \notin \mathcal{C}_N^{(d)}$, then $\sigma y \notin \mathcal{C}_N^{(d)}$. And, if $|y - x| = 1$, then $|\sigma y - \sigma x| = 1$, so

$$(7) \quad \sum_{y' \sim \sigma x} f(y') = \sum_{y \sim x} f(\sigma y).$$

We say $\Omega \subseteq \mathcal{C}_N^{(d)}$ is a fundamental domain if there exists a set $\{\sigma_1, \sigma_2, \dots\} \subset \text{Aut}_{\mathcal{C}_d}$ so that $\bigcup_j \sigma_j \Omega = \mathcal{C}_N^{(d)}$. For example, a fundamental domain of an interval is half of it, while a fundamental domain of a square is a right triangle with one side along a central cross section of the square. The fundamental domains which we consider have coordinate consistency across dimensions. Let $M = \lceil N/2 \rceil$ and

$$(8) \quad \mathcal{S}_M^{(d)} := \{(x_1, \dots, x_d) : M \geq x_1 \geq \dots \geq x_d \geq 1\}.$$

Observe that $\{\mathbf{x}_{d-1} : (\mathbf{x}_{d-1}, 1) \in \mathcal{S}_M^{(d)}\} = \mathcal{S}_M^{(d-1)}$; this is the first step towards proving dimensional reduction on the hypercube.

Let v_t be the odometer function for an initial sandpile, s_0 , on $\mathcal{C}_N^{(d)}$ which is *symmetric*, $\sigma s_0 = s_0$ for all $\sigma \in \text{Aut}_{\mathcal{C}_d}$. We show that the parallel toppling odometer coincides with the *symmetrized* odometer v_t^S on $\mathcal{S}_M^{(d)}$ with appropriate reflecting boundary conditions. That is, for each $x \in \mathcal{S}_M^{(d)}$ and $y \sim x$ there exists a unique rotation or reflection, $\sigma_y \in \text{Aut}_{\mathcal{C}_d}$, so that $\sigma_y y \in \mathcal{S}_M^{(d)}$. Let $y \stackrel{S}{\sim} x$ denote iteration over the set $\{\sigma_y y : y \sim x\}$. For all $t \geq 0$ and $x \in \mathcal{S}_M^{(d)}$, let

$$(9) \quad v_{t+1}^S(x) = \lfloor \frac{s_0(x) + \sum_{y \stackrel{S}{\sim} x} v_t^S(y)}{2d} \rfloor,$$

where $v_0^S \equiv 0$.

We provide an algorithmic construction of this which we use to prove Lemma 2.4 below. Suppose $M \geq 2$, let $x \in \mathcal{S}_M^{(d)}$ be given and define $x_0 = M$ and $x_{d+1} = 1$. The following algorithm produces a sequence of indices describing the symmetrized nearest neighbors of x . Start with $l_0 = 0$ and pick the largest $(d+1) \geq r_0 \geq l_0$ with $x_{l_0} = x_{r_0}$. If $r_0 = (d+1)$, stop, otherwise, set $l_1 = (r_0 + 1)$ and repeat, generating

$$(10) \quad \mathbf{I}^{(M,d)}(x) := \{(l_0, r_0), \dots, (l_n, r_n)\},$$

where $n \leq (d+1)$. Observe that

$$M = x_{l_0} = x_{r_0} < \dots < x_{l_k} = x_{r_k} < \dots < x_{l_n} = x_{r_n} = 1$$

so that

$$(11) \quad \begin{aligned} \sum_{y \stackrel{S}{\sim} x} v_t(y) &= \sum_{k=1}^{n-1} (1 + r_k - l_k) (v_t(x - e_{r_k}) + v_t(x + e_{l_k})) \\ &\quad + (r_0 - l_0) v_t(x - e_{r_0}) \\ &\quad + (r_n - l_n) (v_t(x - e_{r_n}) + v_t(x + e_{l_n})) \end{aligned}$$

where $v_t(x - e_0) = v_t(x + e_{d+1}) := 0$ and

$$(12) \quad v_t(x - e_{r_n}) = \begin{cases} v_t(x) & \text{if } N \text{ is even} \\ v_t(x + e_{l_n}) & \text{if } N \text{ is odd.} \end{cases}$$

Lemma 2.3 (Symmetry). *For each $t \geq 0$ and each $\sigma \in \text{Aut}_{\mathcal{C}_d}$, $\sigma v_t = v_t$. Hence, $v_t = v_t^S$ on $\mathcal{S}_M^{(d)}$.*

Proof. We prove symmetry of v_t by induction and Lemma 2.1. At $t = 0$, $v_t \equiv 0$, so suppose symmetry holds at time t . Let $\sigma \in \text{Aut}_{\mathcal{C}_d}$, $x \in \mathcal{C}_N^{(d)}$ be given. By Lemma 2.1, (7), and the inductive hypothesis,

$$\begin{aligned} v_{t+1}(\sigma x) &= \lfloor \frac{s_0(\sigma x) + \sum_{y' \sim \sigma x} v_t(y')}{2d} \rfloor \\ &= \lfloor \frac{s_0(\sigma x) + \sum_{y \sim x} v_t(\sigma y)}{2d} \rfloor \\ &= \lfloor \frac{s_0(x) + \sum_{y \sim x} v_t(y)}{2d} \rfloor \\ &= v_{t+1}(x). \end{aligned}$$

□

Note that the proof indicates that Lemma 2.3 can be extended in a natural way to other graphs and domains which are preserved under the automorphism group of the graph.

Henceforth, we consider $v_t^{\mathcal{S}}$ in $\mathcal{S}_M^{(d)}$ and drop all $\mathcal{S}$ superscripts. To reduce the number of cases with similar arguments, we only consider $N = 2M$. Indeed, when N is odd, the proofs are identical except for slight changes to the boundary arguments. Also, we will use $\mathcal{S}_M^{(d)}$ to refer exclusively to the sorted fundamental domain of $\mathcal{C}_{2M}^{(d)}$. The expressions $v_t^{(d,M)}$ and $s_t^{(d,M)}$ will refer to the parallel toppling odometers and sandpiles on $\mathcal{S}_M^{(d)}$.

2.5. Derivative comparison. In this section we provide a general parabolic comparison result for first order differences of v_t on $\mathcal{S}_M$ when the initial sandpile, s_0 , is constant. For $w \in \mathbb{Z}^d$, let D_w denote a first order difference operator of the form $D_w v_t(\cdot) = v_t(\cdot) - v_t(\cdot + w)$. Pad v_t by $v_t(x) := 0$ for all $x \notin \mathcal{C}_N$. Denote the interior with respect to w as

$$(13) \quad \text{Int}_w(\mathcal{S}_M) = \{x \in \mathcal{S}_M : y' \in \mathcal{S}_M \text{ for all } |y' - (x + w)| = 1\}$$

and the boundary as

$$(14) \quad \partial_w \mathcal{S}_M = \mathcal{S}_M \setminus \text{Int}_w(\mathcal{S}_M).$$

Observe that every symmetrized $y \sim x \in \mathcal{S}_M$ is of the form $y = (x \pm e_i) \mp d_i$, where d_i is either a reflection, $d_i = e_i$ or a rotation $d_i = e_i - e_j$. We will show that if one can control $D_w v_t$ over the reflecting, rotating, and dissipating boundaries of $\mathcal{S}_M$, then that control persists over time. The dissipating boundary on $\mathcal{S}_M$ is

$$(15) \quad \mathcal{B}_w^{(disp)} \mathcal{S}_M = \{x \in \mathcal{S}_M : (x + w)_i \geq M \text{ for some } 1 \leq i \leq d\}$$

while the reflecting and rotating boundaries are

$$(16) \quad \mathcal{B}_w^{(ref)} \mathcal{S}_M = \{x \in \mathcal{S}_M : (x + w)_i \leq 1 \text{ for some } 1 \leq i \leq d\}$$

and

$$(17) \quad \mathcal{B}_w^{(rot)} \mathcal{S}_M = \{x \in \mathcal{S}_M : (x + w)_i \leq (x + w)_j \text{ for some } 1 \leq i < j \leq d\}.$$

For notational convenience write

$$(18) \quad \mathcal{B}_w = \{\mathcal{B}_w^{(disp)} \cup \mathcal{B}_w^{(ref)} \cup \mathcal{B}_w^{(rot)}\}$$

and $\mathcal{B}_{w_1, \dots, w_n} = \cup_{i=1}^n \mathcal{B}_{w_i}$ for points $w_i \in \mathbb{Z}^d$. Note that in next lemma, we employ our convention to sometimes omit distinguishing sub/superscripts.

Lemma 2.4. *Let $\mathbf{w} := \{w_1, \dots, w_n\}$ be a set of points in $\mathbb{Z}^d$ each equipped with a function $g_j : \mathcal{S} \rightarrow \mathbb{Z}$ which is superharmonic in the interior of $\mathcal{S}$. If*

$$(19) \quad \sup_j (D_{w_j} v_{t_0}(x) - g_j(x)) \leq 0 \quad \text{for all } x \in \mathcal{S}$$

and for all $t \geq t_0$ and $x \in \{\mathcal{B}_0 \cup \mathcal{B}_{\mathbf{w}}\} \mathcal{S}$,

$$(20) \quad \sup_j \left(\sum_{y \sim x} v_t(y) - \sum_{y' \sim (x + w_j)} v_t(y') - 2dg_j(x) \right) \leq 0$$

or

$$\sup_j (D_{w_j} v_{t+1}(x) - g_j(x)) \leq 0$$

then

$$(21) \quad \sup_j (D_{w_j} v_{t+1}(x) - g_j(x)) \leq 0$$

for all $t \geq t_0$ and $x \in \mathcal{S}$.

Proof. We prove this by induction on t , starting at t_0 , the base case guaranteed by (19). Suppose (21) holds at t and let $w_j, x \in \mathcal{S}$ be given. First suppose $x \in \{\text{Int}_{w_j} \cap \text{Int}_0\}(\mathcal{S})$. By Lemma 2.1

$$\begin{aligned} D_{w_j} v_{t+1}(x) - g_j(x) &\leq \lfloor \frac{(2d-1) + \sum_{y \sim x} v_t(y) - \sum_{y' \sim (x+w_j)} v_t(y')}{2d} \rfloor - g_j(x) \\ &= \lfloor \frac{(2d-1) + \sum_{y \sim x} D_{w_j} v_t(y)}{2d} \rfloor - g_j(x) \\ &= \lfloor \frac{(2d-1) + \sum_{y \sim x} (D_{w_j} v_t(y) - g_j(y)) + \sum_{y \sim x} (g_j(y) - g_j(x))}{2d} \rfloor \\ &\leq 0 \end{aligned}$$

as g_j is superharmonic and integer-valued. If $x \in \{\mathcal{B}_0 \cup \mathcal{B}_w\}$, then we either use the same argument or conclude depending on the case in (20). $\square$

As a corollary, we deduce the following discrete quasiconcavity property of v_t on a hypercube, which was proved in [BG07] for axis monotonic initial sandpiles on $\mathbb{Z}^2$. (Note that Aleksanyan and Shahgholian, using a discrete analogue of the method of moving planes, proved axis monotonicity of v_∞ in [AS19].)

Corollary 2.1 (Axis monotonicity [BG07]). *For all $t \geq 1$, $x \in \mathcal{S}$, and all sets of indices*

$$\mathcal{I} = \{i_1, \dots, i_n : 1 \leq i_1 < \dots < i_n \leq d\}$$

and

$$\mathcal{J} = \{j_1, \dots, j_m : i_n < j_1 < \dots < j_m \leq d\}$$

where $n \geq 1$ and $m \geq 0$ we have

$$v_t(x) \geq v_t(x + e_{\mathcal{I}} - e_{\mathcal{J}}) \quad \text{for } (x + e_{\mathcal{I}} - e_{\mathcal{J}}) \in \mathcal{S}$$

where $e_{\mathcal{I}} = \sum_{i \in \mathcal{I}} e_i$ denotes a sum over standard basis vectors indexed by $\mathcal{I}$.

We also have control on the derivative given an odometer upper bound on the dissipating boundary.

Corollary 2.2 (Derivative bound). *Suppose $v_\infty(M, \mathbf{1}_{d-1}) \leq kM$ for integer $k \geq 1$. Then, for all $1 \leq j \leq d$ and $t \geq 0$*

$$(22) \quad v_t(x) - v_t(x + e_j) \leq kx_j.$$

Proof. The claim is immediate if $M = 1$, so suppose $M \geq 2$. Let e_j and x be given and let

$$(23) \quad \mathbf{I}^{(M,d)}(x) := \{(l_0, r_0), \dots, (l_n, r_n)\},$$

be the indices describing the nearest neighbors of x as defined in Section 2.4 above.

Pick the largest index J so that

$$l_J \leq j \leq r_J,$$

$v_t(x + e_j) = v_t(x + e_{l_J})$, and (recalling $l_J = r_{J-1} + 1$)

$$(24) \quad \mathbf{I}^{(M,d)}(x + e_{l_J}) = \begin{cases} \{\dots, (l_{J-1}, r_{J-1} + 1), (l_J + 1, r_J), \dots\} & \text{if } x_{r_{J-1}} = x_{l_J} + 1 \\ \{\dots, (l_{J-1}, r_{J-1}), (l_J, l_J), (l_J + 1, r_J), \dots\} & \text{if } x_{r_{J-1}} > x_{l_J} + 1. \end{cases}$$

As $g_j(x) := kx_j$ is harmonic in the interior of $\mathcal{S}$, it remains to check (20) in Lemma 2.4. For later reference, we label the expression we bound,

$$(25) \quad \sum_{y \sim x} v_t(y) - \sum_{y' \sim (x + e_{l_J})} v_t(y').$$

The computations are similar in other cases, so we assume $x_{r_{J-1}} = x_{l_J} + 1$ and $l_J + 1 \leq r_J$.

Case 1: $J = 0$

As we are on the dissipating boundary, $v_{t+1}(x + e_{l_J}) = 0$ and $x_{l_J} = \dots = x_j = M$, hence

$$v_{t+1}(x) - v_{t+1}(x + e_{l_J}) = v_{t+1}(x) \leq v_\infty(M, \mathbf{1}_{d-1}) \leq kM$$

by axis monotonicity and our assumption on the odometer.

Case 2: $1 < J < n$

We compute (25), observing that all differences except for those near r_J are unaffected by the symmetrization;

$$\begin{aligned} (25) &= \sum_{k=1, k \notin [J-1, J]}^{n-1} (1 + r_k - l_k)(v_t(x - e_{r_k}) - v_t(x - e_{r_k} + e_{l_J})) \\ &\quad + \sum_{k=1, k \notin [J-1, J]}^{n-1} (1 + r_k - l_k)(v_t(x + e_{l_k}) - v_t(x + e_{l_k} + e_{l_J})) \\ &\quad + (r_0 - l_0)(v_t(x - e_{r_0}) - v_t(x - e_{r_0} + e_{l_J})) \\ &\quad + (r_n - l_n)(v_t(x) - v_t(x + e_{l_J}) + v_t(x + e_{l_n}) - v_t(x + e_{l_n} + e_{l_J})) \\ &\quad + \star_{\{J-1, J\}} \\ &\leq (2d - 2(1 + r_J - l_{J-1}))kx_j + \star_{\{J-1, J\}} \end{aligned}$$

where $\star_{\{J-1, J\}}$ is defined as the sum of terms in the difference with indices $\{J-1, J\}$. This can then be computed,

$$\begin{aligned} \star_{\{J-1, J\}} &= (1 + r_J - l_J)(v_t(x - e_{r_J}) + v_t(x + e_{l_J})) \\ &\quad - (r_J - l_J)(v_t(x + e_{l_J} - e_{r_J}) + v_t(x + e_{l_J} + e_{l_{J+1}})) \\ &\quad + (1 + r_{J-1} - l_{J-1})(v_t(x - e_{r_{J-1}}) + v_t(x + e_{l_{J-1}})) \\ &\quad - (2 + r_{J-1} - l_{J-1})(v_t(x + e_{l_J} - e_{l_J}) + v_t(x + e_{l_J} + e_{l_{J-1}})) \\ &\leq (r_J - l_J)2kx_j + (1 + r_{J-1} - l_{J-1})(k(x_j + 1 - 1) + kx_j) \\ &\quad + k(x_j - 1) + k(x_j + 1) \\ &= 2(1 + r_J - l_{J-1})kx_j. \end{aligned}$$

Case 3: $J = 1 < n$

We bound differences with indices $\{0, 1\}$ in (25),

$$\begin{aligned}
\star_{\{0,1\}} &\leq (r_0 - l_0)kx_{l_1} \\
&\quad + (r_1 - l_1)2kx_{l_1} \\
&\quad + (v_t(x - e_{r_1}) - v_t(x)) + (v_t(x + e_{l_1}) - 0) \\
&\leq (r_0 - l_0)kx_{l_1} + (r_1 - l_1)2kx_{l_1} + k(x_{l_1} - 1) + k(x_{l_1} + 1) \\
&\leq 2(r_1 - l_0)kx_{l_1}.
\end{aligned}$$

Case 4: $J = n > 1$

We bound differences with indices $\{n-1, n\}$ in (25),

$$\begin{aligned}
\star_{\{n-1,n\}} &\leq (1 + r_{n-1} - l_{n-1})(k(x_{r_{n-1}} - 1) + kx_{l_n}) \\
&\quad + (r_n - l_n - 1)(kx_{l_n} + kx_{l_n+1}) \\
&\quad + (v_t(x) - v_t(x)) + (v_t(x + e_{l_n}) - v_t(x + e_{l_n} + e_{l_{n-1}})) \\
&\leq (1 + r_{n-1} - l_{n-1})2kx_{l_n} + (r_n - l_n - 1)2kx_{l_n} \\
&\quad + k(x_{l_{n-1}}) \\
&\leq 2(r_n - l_{n-1})kx_{l_n}.
\end{aligned}$$

In the last step we used $x_{l_{n-1}} = x_{l_n} + 1 = 2x_{l_n}$.

Case 5: $J = n = 1$

We bound differences with indices $\{0, n\}$ in (25),

$$\begin{aligned}
\star_{\{0,n\}} &\leq (r_0 - l_0)(k(x_{r_0} - 1)) \\
&\quad + (r_n - l_n - 1)(kx_{l_n} + kx_{l_n+1}) \\
&\quad + (v_t(x + e_{l_n}) - 0) + (v_t(x) - v_t(x)) \\
&\leq (r_0 - l_0)2kx_{l_n} + (r_n - l_n - 1)2kx_{l_n} \\
&\quad + kx_{l_n} \\
&\leq 2(r_n - l_0 - 1)kx_{l_n}.
\end{aligned}$$

□

2.6. Weak topple control. We provide a difference-in-time analogue of Lemma 2.4

Lemma 2.5. *For all $t \geq t_0$ and $j \geq 0$,*

$$\max_{z \in \mathcal{S}} (v_{t+j}(z) - v_t(z)) \leq \max_{z \in \mathcal{S}} (v_{t_0+j}(z) - v_{t_0}(z))$$

Proof. We induct on t starting at t_0 . Suppose the result holds for $(t-1)$ and let $x \in \mathcal{S}$ be given. Lemma 2.1 implies

$$v_{t+j}(x) - v_t(x) \leq \left\lfloor \frac{(2d-1) + \sum_{y \sim x} (v_{t+j-1}(y) - v_{t-1}(y))}{2d} \right\rfloor,$$

hence, by induction

$$v_{t+j}(x) - v_t(x) \leq \left\lfloor \frac{(2d-1) + 2d(\max_{z \in \mathcal{S}} (v_{t_0+j}(z) - v_{t_0}(z)))}{2d} \right\rfloor = \max_{z \in \mathcal{S}} (v_{t_0+j}(z) - v_{t_0}(z)).$$

□

3. EXPLICIT SOLUTIONS WHEN $M \leq 2$

In this section, we compute $v_t^{(d)}$ when $s_0^{(d)} \equiv 2d + (d-1)$ for all $d \geq 1$ when $M = 2$. We also show that dimensional reduction does not occur at dimension $d = d_0$ when $s_0^{(d)} \equiv 2d + (d_0 - 1)$ and $M = 1$.

3.1. $M = 1$. As we do not know how to define a 0-dimensional sandpile, suppose $d_0 \geq 2$.

Proposition 3.1. *When $s_0^{(d)} \equiv 2d + (d_0 - 1)$, $v_\infty^{(d_0)} \equiv 1$ but $v_\infty^{(d_0-1)} \equiv 2$.*

Proof. In dimension d , a corner site of the hypercube has d internal neighbors so $\Delta^{(d)}(\mathbf{1}) = -2d + d = -d$. Hence, in dimension d_0 ,

$$s_1^{(d_0)}(\mathbf{1}) = (2d_0 + (d_0 - 1)) - d_0 = 2d_0 - 1$$

however, in dimension $(d_0 - 1)$,

$$s_1^{(d_0-1)}(\mathbf{1}) = (2(d_0 - 1) + (d_0 - 1)) - (d_0 - 1) = 2(d_0 - 1).$$

□

3.2. $M = 2$. Now, suppose $d \geq 1$. After a radial reparameterization of $\mathcal{S}_2$, arbitrary dimensional sandpiles become one-dimensional with a simple nearest-neighbor toppling rule. Indeed, every $\mathbf{x} \in \mathcal{S}_2$ is of the form $\mathbf{x} = (\mathbf{2}_x, \mathbf{1}_{d-x})$, for $x = 0, \dots, d$. Overload notation and consider s_t and v_t as functions on $\{0, \dots, d\}$. The Laplacian on the one-dimensional graph can then be computed using symmetry.

Lemma 3.1. *If we define $0 = v_t(d+1) = v_t(-1)$, then*

$$\Delta v_t(x) = (-d - x)v_t(x) + (d - x)v_t(x+1) + xv_t(x-1).$$

Proof. Let $\mathbf{x} = (\mathbf{2}_x, \mathbf{1}_{d-x})$ so that $\mathbf{I}^{(2,d)}(\mathbf{x}) = \{(0, x), (x+1, d+1)\}$. Hence, by definition of the symmetric Laplacian,

$$\begin{aligned} \Delta v_t(\mathbf{x}) &= -2dv_t(\mathbf{x}) + xv_t(\mathbf{x} - e_x) + (d - x)(v_t(\mathbf{x}) + v_t(\mathbf{x} + e_{x+1})) \\ &= -2dv_t(x) + (d - x)v_t(x) + (d - x)v_t(x+1) + xv_t(x-1). \end{aligned}$$

□

See Figure 4 for a display of the odometer throughout the parallel toppling process when $s_0^{(d)} \equiv 2d + (d-1)$ in dimension $d = 100$. Visually, a contiguous block of decreasing size fires at each step, followed by a ripple of outwards firings. For $t > t_d := (\lceil \sqrt{d-1} \rceil + 1)$, the firing block appears to decrease by one every step. In particular, if a_t indexes the right edge of the block at time t , then $a_1 = d$ and

$$a_t = \begin{cases} \lfloor \frac{d-1}{t-1} \rfloor & \text{for } t \leq t_d \\ a_{t-1} - 1 & \text{for } t > t_d. \end{cases}$$

This leads to a simple formula for v_t .

Proposition 3.2. *For all $d \geq 1$, when $s_0^{(d)} = 2d + (d-1)$, the radially reparameterized parallel toppling odometer has the following form. For all $t \geq 1$,*

$$(26) \quad v_t(x) = \begin{cases} v_{t-1}(x) + 1 & \text{for } x \leq a_t \\ v_{t-1}(x) & \text{for } a_t < x \leq a_{t-1}. \end{cases}$$

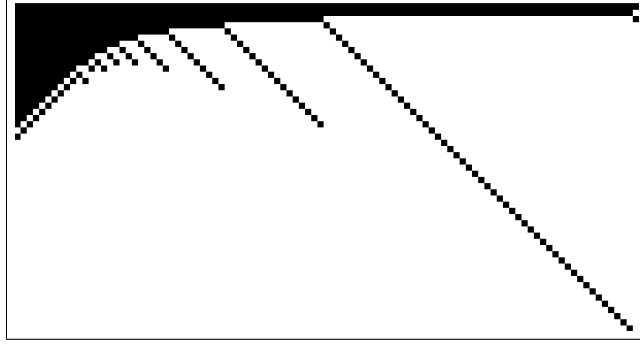


FIGURE 4. The parallel toppling odometer for $s_0 \equiv 2d + (d - 1)$ when $d = 100$ and $M = 2$. A black pixel in row t and column x indicates that $v_t(x) = v_{t-1}(x) + 1$. The top row is $t = 0$ and t increases from top to bottom. The left column is $x = 1$ and x increases from left to right.

And for each $t' < t$ and $a_{t'-1} \geq x > a_{t'}$

$$(27) \quad v_t(x) = v_{t-1}(x - 1).$$

Proof. We induct on t . Since $s_0 \geq 2d$, $v_1 \equiv 1$. Suppose (26) and (27) hold for all $t' \leq t$.

Step 1: (26)

By strong induction for $t' \leq t$, (26) implies $v_t(x) = t$ for $x \leq a_t$. Thus,

$$s_t(x) = \begin{cases} 2d + (d - 1) - tx & \text{for } x < a_t \\ 2d + (d - 1) - tx - (d - x) & \text{for } x = a_t. \end{cases}$$

Let $g(x) := (d - 1) - tx$. If $g(x) \geq 0$, $v_{t+1}(x) = v_t(x) + 1$, otherwise $v_{t+1}(x) = v_t(x)$. When $a_{t+1} < x \leq a_t$, $g(x) < 0$. Indeed, $g(a_{t+1}) \leq (t - 1)$ and $g(x + 1) - g(x) = -t$.

As g is increasing in x , it remains to check $g(a_{t+1}) \geq 0$ for all t . If $x \leq \frac{d-1}{t}$ then $g(x) \geq 0$. If $(t + 1) > t_d$, then

$$\frac{d-1}{t-1} - \frac{d-1}{t} \leq \frac{d-1}{\sqrt{d-1}(\sqrt{d-1}+1)} \leq 1,$$

thus

$$\begin{aligned} a_{t+1} &= a_t - 1 \\ &\leq \frac{d-1}{t-1} - 1 \\ &\leq \frac{d-1}{t}. \end{aligned}$$

Step 2: (27)

Now, take $a_{t'-1} \geq x > a_{t'}$ for $1 \leq t' \leq (t - 1)$. If $v_t(x - 1) = v_t(x) + 1$, then by strong induction for $t' \leq t$, (27) and (26) imply that $v_t(x - 2) = v_t(x - 1) = t'$ and

$v_t(x) = t' - 1 = v_t(x + 1)$. Thus,

$$\begin{aligned} s_t(x) &\geq 2d + (d - 1) - (t' - 2)x \\ &\geq 2d + (d - 1) - (t' - 2)a_{t'-1} \\ &\geq 2d + (d - 1) - (d - 1) \\ &= 2d. \end{aligned}$$

However,

$$\begin{aligned} s_t(x - 1) &= 2d + (d - 1) - t'x - d + x \\ &= 2d - 1 - (t' - 1)x \\ &\leq 2d - 1, \end{aligned}$$

as $t' \geq 1$. If $v_t(x - 1) = v_t(x) = v_t(x + 1) = (t' - 1)$, then

$$\begin{aligned} s_t(x) &\leq 2d + (d - 1) - (t' - 1)x \\ &< 2d + (d - 1) - (t' - 1)(a_{t'}) \\ &\leq 2d. \end{aligned}$$

□

4. ODOMETER REGULARITY WHEN $d = 1$

From here onward, suppose $s_0 \equiv 2d$. We start the inductive proof of Theorem 1.1 by establishing some regularity of the odometer in the critical dimension $d = d_0 = 1$. In the next section, we inductively use dimensional reduction to show that $d \geq 2$ sandpiles inherit this regularity. This regularity ensures that the dynamics of lower-dimensional sandpiles agree with their higher-dimensional embeddings.

When reading Section 5 below, the reader should observe that whenever Proposition 4.1 (or something close to it) holds, dimensional reduction follows. For example, if a version of this result is established in every critical dimension $d_0 \geq 1$, then dimensional reduction follows for all sandpiles of the form $2d + (d_0 - 1)$ in dimensions $d > d_0$. Proposition 3.2 should be understood as a step in this direction.

Proposition 4.1. *Recall the definition of τ_j from Theorem 1.1. For all $M \geq 2$ and $d \geq 1$, the odometer maintains the following properties throughout the parallel toppling process.*

Self-similarity: *For each $1 \leq j \leq M$ and $t \leq \tau_j$*

$$(28) \quad v_t^{(M)}(\mathbf{x}) = v_t^{(j)}(\mathbf{x} - (M - j)) \text{ for } \mathbf{x} > M - j.$$

Weak facet compatibility: *For all $\mathbf{x}_i \geq 2$, $t \geq 1$, $i \geq 0$, $j \geq 0$ and $i + j + 1 = d$*

$$\begin{aligned} (29) \quad v_t^{(M)}(\mathbf{x}_i, 1, \mathbf{1}_j) &= v_t^{(M)}(\mathbf{x}_i, 2, \mathbf{1}_j) + 1 \\ &\implies \\ v_{t+1}^{(M)}(\mathbf{x}_i, 1, \mathbf{1}_j) &= v_{t+1}^{(M)}(\mathbf{x}_i, 1, \mathbf{1}_j). \end{aligned}$$

Strong facet compatibility: *For all $\mathbf{x}_i \geq 2$, $j \geq 0$, $i < d$, ($t < \tau_M$ and $i \geq 0$) or ($i \geq 1$ and $t \geq \tau_M$)*

$$(30) \quad v_t^{(M)}(\mathbf{x}) - v_t^{(M)}(\mathbf{x} + 2e_{i+1}) \leq 2$$

and

$$\begin{aligned}
 v_t^{(M)}(\mathbf{x}_i, 1, \mathbf{1}_j) &= v_t^{(M)}(\mathbf{x}_i, 2, \mathbf{1}) + 1 \\
 &\implies \\
 (31) \quad v_{t+1}^{(M)}(\mathbf{x}_i, 1, \mathbf{1}_j) &= v_t^{(M)}(\mathbf{x}_i, 1, \mathbf{1}_j) \\
 v_{t+1}^{(M)}(\mathbf{x}_i, 2, \mathbf{1}_j) &= v_t^{(M)}(\mathbf{x}_i, 2, \mathbf{1}_j) + 1.
 \end{aligned}$$

Strong topple control: For all $t \geq \tau_{M-1}$,

$$(32) \quad \sup_{\mathbf{x} \in \mathcal{S}_M} (v_{t+2}(\mathbf{x}) - v_t(\mathbf{x})) \leq 1.$$

Proof of Proposition 4.1 for $d = d_0 = 1$. The proof proceeds by induction on M and then on t . When $M = 2$,

	$v_t^{(M)}(1)$	$v_t^{(M)}(2)$
$t = 1$	1	1
$t = 2$	2	1
$t = 3$	2	2
$t = 4$	3	2

and $v_1^{(M-1)}(1) = v_2^{(M-1)}(1) = 1$ which verifies the base case. Now, let M be given and note that $v_1^{(M)}(x) = 1$ for all $1 \leq x \leq M$ and $v_2^{(M)}(x) = 2$ for $1 \leq x < M$ and $v_2^{(M)}(M) = 1$. Hence, suppose (28),(29),(30),(31), hold for $(M-1)$ for all $t \geq 1$ and suppose they hold for M for all $t' \leq (t-1)$. We verify each inductive step.

Self-similarity: (28). By strong induction, it suffices to show $v_t^{(M)}(x) = v_t^{(M-1)}(x-1)$ for $x \geq 2$. By Lemma 2.1,

$$v_t^{(M)}(x) = 1 + \lfloor \frac{v_{t-1}^{(M)}(x+1) + v_{t-1}^{(M)}(x-1)}{2} \rfloor$$

for $x > 1$. Hence, by (28) at $(t-1)$, for $x > 2$,

$$v_t^{(M)}(x) = 1 + \lfloor \frac{v_{t-1}^{(M-1)}(x) + v_{t-1}^{(M-1)}(x-2)}{2} \rfloor = v_t^{(M-1)}(x-1).$$

For $x = 1$, we have reflection at the origin,

$$v_t^{(M-1)}(1) = 1 + \lfloor \frac{v_{t-1}^{(M-1)}(2) + v_{t-1}^{(M-1)}(1)}{2} \rfloor = 1 + \lfloor \frac{v_{t-1}^{(M)}(3) + v_{t-1}^{(M)}(2)}{2} \rfloor.$$

Hence, if $v_{t-1}^{(M)}(1) = v_{t-1}^{(M)}(2)$, then $v_t^{(M-1)}(1) = v_t^{(M)}(2)$.

When $v_{t-1}^{(M)}(1) = v_{t-1}^{(M)}(2) + 1$, we instead use strong facet compatibility in both layers. If $v_{t-1}^{(M-1)}(1) = v_{t-1}^{(M-1)}(2)$, then $v_t^{(M-1)}(1) = v_{t-1}^{(M-1)}(1) + 1$ and we are done, so suppose not. Since sites topple at most once per time step by Lemma 2.5, the odometer must then be, for some $v \geq 2$:

	$v^{(M)}(1)$	$v^{(M)}(2)$	$v^{(M)}(3)$
$t-2$	v	v	$v-1$
$t-1$	$v+1$	v	$v-1$
t	$v+1$	$v+1$	$\geq (v-1)$

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This contradicts strong facet compatibility for $v_t^{(M-1)}(1)$ from $(t-2) \rightarrow (t-1)$, which we can use as $t \leq \tau_{M-1}$ and hence $(t-1) < \tau_{M-1}$.

Weak facet compatibility: (29). If $v_t(1) = v_t(2) + 1$, then $\Delta v_t(1) = -1$ and so $v_{t+1}(1) = v_t(1)$.

Strong facet compatibility: (30) and (31). We use Lemma 2.4 to show (30). The function $g(x) = 2x$ is harmonic in the interior of the interval so it suffices to check the dissipating and reflecting boundaries. We control the dissipating boundary using $t < \tau_M$ and the reflecting boundary with (29).

As $t < \tau_M$, $v_t(M) \leq (M-1)$ and hence by Corollary 2.2, $v_t(M-1) \leq v_t(M) + (M-1) \leq 2(M-1)$. For the reflecting boundary, i.e, $x = 1$, we check

$$\sum_{y \sim x} v_t(y) - \sum_{y' \sim (x+2)} v_t(y) = (v_t(1) - v_t(2)) + (v_t(2) - v_t(4)) \leq (v_t(1) - v_t(2)) + 4.$$

If $v_t(1) - v_t(2) = 1$, then $v_{t+1}(1) = v_t(1)$ by weak facet compatibility. Otherwise, $\sum_{y \sim x} v_t(y) - \sum_{y' \sim (x+2)} v_t(y) \leq 4$ and we conclude that

$$v_t(x) - v_t(x+2) \leq 2x.$$

Taking $x = 1$, this implies

$$\Delta v_t(2) \geq -2v_t(2) + v_t(1) + v_t(1) - 2 \geq 0,$$

which shows (31).

Strong topple control: (32). By Lemma 2.5, it suffices to show

$$\sup_{x \in \mathcal{S}_M} (v_{\tau_{M-1}+2}(x) - v_{\tau_{M-1}}(x)) \leq 1$$

First observe that (28) for $v_t^{(M)}$ and (32) for $v_t^{(M-1)}$ imply that

$$(33) \quad \tau_{M-1} \geq \tau_{M-2} + 2.$$

Suppose for sake of contradiction that

$$(v_{\tau_{M-1}+2}(x) - v_{\tau_{M-1}}(x)) = 2$$

for some $1 \leq x \leq M$. Lemma 2.5 then implies that some neighbor $y \sim x$ must have toppled twice previously. Pick the maximal such y . We consider three cases for y .

Case 1: $y \geq 3$

We first note that $v_{\tau_{M-1}+1}^{(M)}(y) = v_{\tau_{M-1}+1}^{(M-1)}(y-1)$. Indeed, by (28), as $(y-1) \geq 2$,

$$\begin{aligned} v_{\tau_{M-1}+1}^{(M)}(y) &= 1 + \left\lfloor \frac{v_{\tau_{M-1}}^{(M)}(y+1) + v_{\tau_{M-1}}^{(M)}(y-1)}{2} \right\rfloor \\ &= 1 + \left\lfloor \frac{v_{\tau_{M-1}}^{(M-1)}(y) + v_{\tau_{M-1}}^{(M-1)}(y-2)}{2} \right\rfloor \\ &= v_{\tau_{M-1}+1}^{(M-1)}(y-1) \end{aligned}$$

Hence,

$$2 = v_{\tau_{M-1}+1}^{(M)}(y) - v_{\tau_{M-1}-1}^{(M)}(y) = v_{\tau_{M-1}+1}^{(M-1)}(y-1) - v_{\tau_{M-1}-1}^{(M-1)}(y-1),$$

which contradicts (32) for $v_t^{(M-1)}$.

Case 2: $y = 2$

We claim that $v_{\tau_{M-1}+1}^{(M)}(2) = v_{\tau_{M-1}+1}^{(M-1)}(1)$, in which case we can use the argument of Case 1. If not, then $v_{\tau_{M-1}}^{(M)}(2) = v_{\tau_{M-1}}^{(M)}(3) + 1$ but $v_{\tau_{M-1}}^{(M)}(1) = v_{\tau_{M-1}}^{(M)}(2) + 1$. This implies that either

$$v_{\tau_{M-1}-1}^{(M)}(1) = v_{\tau_{M-1}-1}^{(M)}(2) + 2$$

or

$$v_{\tau_{M-1}-1}^{(M)}(1) = v_{\tau_{M-1}-1}^{(M)}(2) + 1$$

and

$$v_{\tau_{M-1}}^{(M)}(1) = v_{\tau_{M-1}-1}^{(M)}(1) + 1$$

both which contradict weak facet compatibility.

Case 3: $y = 1$

In this case, the odometer near the center must be, for some $v \geq 1$,

	$v_{\tau_{M-1}-1}^{(M)}(1)$	$v_{\tau_{M-1}}^{(M)}(2)$	$v_{\tau_{M-1}+1}^{(M)}(3)$
$\tau_{M-1} - 1$	v	v	v
τ_{M-1}	$v + 1$	$v + 1$	v
$\tau_{M-1} + 1$	$v + 2$	$v + 1$	$\geq (v)$

This shows $v_{\tau_{M-1}-2}^{(M)}(2) = v - 1$. Indeed, if $v_{\tau_{M-1}-2}^{(M)}(1) = v - 1$, then as $\Delta v_{\tau_{M-1}-2}^{(M)}(1) \geq 0$ $v_{\tau_{M-1}-2}^{(M)}(2) = v - 1$. If $v_{\tau_{M-1}-2}^{(M)}(1) = v$, then $\Delta v_{\tau_{M-1}-2}^{(M)}(1) \leq -1$ and $v_{\tau_{M-1}-2}^{(M)}(2) = v - 1$. Hence,

$$v_{\tau_{M-1}}^{(M-1)}(1) = v_{\tau_{M-1}}^{(M)}(2) = v_{\tau_{M-1}-2}^{(M)}(2) + 2 = v_{\tau_{M-1}-2}^{(M-1)}(2) + 2,$$

which contradicts (32) for $v_t^{(M-1)}$ using (33). □

Note that the comparison principle for sandpiles (see, for example, Proposition 3.3 in [BR21]) shows

$$v_\infty(x) = \frac{1}{2} (M(M+1) - x(x-1)),$$

and so $v_\infty(x) - v_\infty(x+1) = x$. Hence we must use an assumption like $t < \tau_M$ for strong facet compatibility.

5. ODOMETER REGULARITY AND DIMENSIONAL REDUCTION

We now prove Proposition 4.1 for $d \geq 2$ together with dimensional reduction,

$$(34) \quad v_t^{(d,M)}(\mathbf{x}_{d-1}, 1) = v_t^{(d-1,M)}(\mathbf{x}_{d-1}),$$

by strong induction on M , d , and t . Specifically, given M , d , and t , suppose

$$(28), (29), (30), (31), (32)$$

hold for

$$\begin{aligned} v_{t'}^{(d',M')} & \text{ for all } M' \geq 1, t' \geq 1, d' < d, \\ v_{t'}^{(d,M')} & \text{ for all } M' < M, t' \geq 1, \\ v_{t'}^{(d,M)} & \text{ for all } t' < t. \end{aligned}$$

We also suppose (34) holds for $v_{t-1}^{(d',M')}$ for all $d' \geq 2$ and $M' \leq M$. Indeed, $v_1^{(d,M)} \equiv 1$ for all $d \geq 1$ and $M \geq 1$.

5.1. Dimensional reduction. We start the induction in time by proving dimensional reduction given odometer regularity at $(t-1)$. Let $\mathbf{x}_{d-1}$ be given and pick the smallest $d > i \geq 0$ so that $(\mathbf{x}_i, \mathbf{1}_{d-i}) = (\mathbf{x}_{d-1}, 1)$. By symmetry,

$$\begin{aligned} \Delta^{(d)} v_{t-1}^{(d)}(\mathbf{x}_i, \mathbf{1}_{d-i}) &= -2d v_{t-1}^{(d)}(\mathbf{x}_i, \mathbf{1}_{d-i}) \\ &\quad + \sum_{\mathbf{y}_i \sim \mathbf{x}_i} v_{t-1}^{(d)}(\mathbf{y}_i, \mathbf{1}_{d-i}) \\ &\quad + (d-i) \left(v_{t-1}^{(d)}(\mathbf{x}_i, \mathbf{1}_{d-i}) + v_{t-1}^{(d)}(\mathbf{x}_i, 2, \mathbf{1}_{d-i-1}) \right). \end{aligned}$$

We consider two cases at time $(t-1)$.

Case 1: $v_{t-1}^{(d)}(\mathbf{x}_i, \mathbf{1}_{d-i}) = v_{t-1}^{(d)}(\mathbf{x}_i, 2, \mathbf{1}_{d-i-1})$

By (34) at $(t-1)$, $v_{t-1}^{(d)}(\mathbf{x}_i, \mathbf{1}_{d-i}) = v_{t-1}^{(i+1)}(\mathbf{x}_i, 1)$. Thus,

$$\begin{aligned} \Delta^{(d)} v_{t-1}^{(d,M)}(\mathbf{x}_i, \mathbf{1}_{d-i}) &= -2i v_{t-1}^{(i+1)}(\mathbf{x}_i, 1) + \sum_{\mathbf{y}_i \sim \mathbf{x}_i} v_{t-1}^{(i+1)}(\mathbf{y}_i, 1) \\ &= \Delta^{(i+1)} v_{t-1}^{(i+1,M)}(\mathbf{x}_i, 1) \end{aligned}$$

which concludes this case as $v_t^{(d)} = v_{t-1}^{(d)} + 1$ ($\Delta^{(d)} v_{t-1}^{(d)} \geq 0$).

Case 2: $v_{t-1}^{(d)}(\mathbf{x}_i, \mathbf{1}_{d-i}) = v_{t-1}^{(d)}(\mathbf{x}_i, 2, \mathbf{1}_{d-i-1}) + 1$

If $i \leq (d-2)$, then (29) for $(t-1) \rightarrow t$ for both $v_{t-1}^{(i+1)}$ and $v_{t-1}^{(d)}$ imply that

$$v_{t-1}^{(i+1)}(\mathbf{x}_i, 1) = v_{t-1}^{(d)}(\mathbf{x}_i, \mathbf{1}_{d-i}) = v_t^{(d)}(\mathbf{x}_i, \mathbf{1}_{d-i}) = v_t^{(i+1)}(\mathbf{x}_i, 1).$$

If $i = (d-1)$, then (31) and (34) at $(t-1)$ and $(t-2)$ imply that

$$v_{t-1}^{(d-1)}(\mathbf{x}_{d-1}) = v_{t-1}^{(d)}(\mathbf{x}_{d-1}, 1) = v_{t-2}^{(d)}(\mathbf{x}_{d-1}, 1) + 1 = v_{t-2}^{(d-1)}(\mathbf{x}_{d-1}) + 1.$$

If $(t-2) \geq \tau_{M-1}$, (32) for $v_t^{(i)}$ and $v_t^{(d)}$ imply that

$$v_t^{(d-1)}(\mathbf{x}_{d-1}) = v_{t-1}^{(d-1)}(\mathbf{x}_{d-1}) = v_{t-1}^{(d)}(\mathbf{x}_{d-1}, 1) = v_t^{(d)}(\mathbf{x}_{d-1}, 1).$$

If $(t-2) < \tau_{M-1}$, then (28) and (34) for $v_{t-1}^{(M-1)}$ show

$$v_{t-1}^{(d,M)}(\mathbf{x}_{d-1}, 2) = v_{t-1}^{(d,M-1)}(\mathbf{x}_{d-1} - 1, 1) = v_{t-1}^{(d-1,M-1)}(\mathbf{x}_{d-1} - 1).$$

Similarly,

$$v_{t-1}^{(d,M)}(\mathbf{x}_{d-1}, 1) = v_{t-1}^{(d-1,M)}(\mathbf{x}_{d-1}) = v_{t-1}^{(d-1,M-1)}(\mathbf{x}_{d-1} - 1).$$

Therefore, for all $t' \leq \tau_{M-1}$,

$$(35) \quad v_{t'}^{(d,M)}(\mathbf{x}_{d-1}, 2) = v_{t'}^{(d,M)}(\mathbf{x}_{d-1}, 1),$$

this however contradicts the case we are in.

5.2. Odometer regularity for $d \geq 2$. We verify each inductive step.

Self-similarity: (28). As (28) holds for $(M-1)$ at t , it suffices to show that if $t \leq \tau_{M-1}$ and $\mathbf{x} > 1$,

$$(36) \quad v_t^{(M)}(\mathbf{x}) = v_t^{(M-1)}(\mathbf{x} - 1).$$

We split verification of this into cases.

Case 1: $\mathbf{x} > 2$

As (36) holds for $(t-1)$, by Lemma 2.1,

$$\begin{aligned} v_t^{(M)}(\mathbf{x}) &= \lfloor \frac{s_0(\mathbf{x}) + \sum_{\mathbf{y} \sim \mathbf{x}} v_{t-1}^{(M)}(\mathbf{y})}{2d} \rfloor \\ &= \lfloor \frac{s_0(\mathbf{x} - 1) + \sum_{\mathbf{y} \sim (\mathbf{x} - 1)} v_{t-1}^{(M-1)}(\mathbf{y})}{2d} \rfloor \\ &= v_t^{(M-1)}(\mathbf{x} - 1). \end{aligned}$$

Case 2: $\mathbf{x} = (\mathbf{x}_j, 2)$ for $\mathbf{x}_j > 2$ and $0 \leq j < d$

We show that if $\Delta v_{t-1}^{(M)}(\mathbf{x}) \geq 0$, then $\Delta v_{t-1}^{(M-1)}(\mathbf{x} - 1) \geq 0$. First, decompose the Laplacian into a sum of discrete second differences,

$$\Delta v_{t-1}^{(M)}(\mathbf{x}) = \sum_{i=1}^d \Delta_{(i)} v_{t-1}^{(M)}(\mathbf{x}),$$

where

$$\Delta_{(i)} v_{t-1}^{(M)}(\mathbf{x}) = -2v_{t-1}^{(M)}(\mathbf{x}) + v_{t-1}^{(M)}(\mathbf{x} + e_i) + v_{t-1}^{(M)}(\mathbf{x} - e_i).$$

Observe that (28) at $(t-1)$ implies, $\Delta_{(i)} v_{t-1}^{(M)}(\mathbf{x}) = \Delta_{(i)} v_{t-1}^{(M-1)}(\mathbf{x} - 1)$ for all $i \leq j$ and for $i > j$,

$$\Delta_{(i)}(v_{t-1}^{(M-1)}(\mathbf{x} - 1) - v_{t-1}^{(M)}(\mathbf{x})) = v_{t-1}^{(M-1)}(\mathbf{x} - e_i - 1) - v_{t-1}^{(M)}(\mathbf{x} - e_i).$$

By reflectional symmetry, for each $i > j$,

$$v_{t-1}^{(M-1)}(\mathbf{x} - e_i - 1) = v_{t-1}^{(M-1)}(\mathbf{x} - 1) = v_{t-1}^{(M)}(\mathbf{x}),$$

thus

$$(37) \quad \Delta_{(i)}(v_{t-1}^{(M-1)}(\mathbf{x} - 1) - v_{t-1}^{(M)}(\mathbf{x})) = v_{t-1}^{(M)}(\mathbf{x}) - v_{t-1}^{(M)}(\mathbf{x} - e_i).$$

If $v_{t-1}^{(M)}(\mathbf{x}) = v_{t-1}^{(M)}(\mathbf{x} - e_i)$ for all $i > j$, we are done, so suppose otherwise.

Take $i > j$ where (37) $\neq 0$. By (31),

$$(38) \quad v_{t-2}^{(M)}(\mathbf{x}) = v_{t-1}^{(M)}(\mathbf{x}) = v_{t-1}^{(M)}(\mathbf{x} - e_i) - 1 = v_{t-2}^{(M)}(\mathbf{x} - e_i).$$

By (28) and (31) for $v_{t-2}^{(M-1)}$, if

$$v_{t-1}^{(M-1)}(\mathbf{x} - 1) = v_{t-1}^{(M-1)}(\mathbf{x} - 1 + e_i) + 1$$

then

$$v_{t-1}^{(M)}(\mathbf{x}) = v_{t-1}^{(M-1)}(\mathbf{x} - 1) = v_{t-2}^{(M-1)}(\mathbf{x} - 1) + 1 = v_{t-2}^{(M)}(\mathbf{x}) + 1$$

which contradicts (38). Moreover, by (31) we must have for each neighbor $(\mathbf{y} - e_i) \sim (\mathbf{x} - e_i)$, $v_{t-1}^{(M)}(\mathbf{y}) \geq v_{t-2}^{(M)}(\mathbf{y} - e_i)$. Thus,

$$(39) \quad \Delta v_{t-1}^{(M-1)}(\mathbf{x} - 1) \geq \Delta v_{t-2}^{(M)}(\mathbf{x} - e_i) \geq 0.$$

Strong facet compatibility: (30) and (31). We first use

$$(40) \quad v_t(\mathbf{x}_{i-1}, 1, \mathbf{1}_j) - v_t(\mathbf{x}_{i-1}, 3, \mathbf{1}_j) \leq 2$$

together with the inductive hypotheses to show (31), then we verify (30) below.

Suppose $v_t(\mathbf{x}_{i-1}, 1, \mathbf{1}_j) = v_t(\mathbf{x}_{i-1}, 2, \mathbf{1}_j) + 1$. By (31) at $(\mathbf{x}_i, 1, \mathbf{1}_j)$ from $(t-1) \rightarrow t$, $\Delta v_{t-1}(\mathbf{x}_i, 1, \mathbf{1}_j) \geq 0$. Hence, it suffices to show

$$\Delta v_t(\mathbf{x}_i, 2, \mathbf{1}_j) \geq \Delta v_{t-1}(\mathbf{x}_i, 1, \mathbf{1}_j).$$

We use symmetry to decompose each Laplacian;

$$\begin{aligned} \Delta v_t(\mathbf{x}_{i-1}, 2, \mathbf{1}_j) &= -2dv_t(\mathbf{x}_{i-1}, 2, \mathbf{1}_j) \\ (41) \quad &+ \sum_{j'=1}^{(i-1)} (v_t(\mathbf{x}_i + e_{j'}, 2, \mathbf{1}_j) + v_t(\mathbf{x}_i - e_{j'}, 2, \mathbf{1}_j)) \\ (42) \quad &+ v_t(\mathbf{x}_i, 1, \mathbf{1}_j) + v_t(\mathbf{x}_i, 3, \mathbf{1}_j) \\ (43) \quad &+ j(v_t(\mathbf{x}_i, 2, \mathbf{1}_j) + v_t(\mathbf{x}_i, 2, 2, \mathbf{1}_{j-1})) \end{aligned}$$

while

$$\begin{aligned} \Delta v_{t-1}(\mathbf{x}_{i-1}, 1, \mathbf{1}_j) &= -2dv_{t-1}(\mathbf{x}_{i-1}, 1, \mathbf{1}_j) \\ (44) \quad &+ \sum_{j'=1}^{(i-1)} (v_{t-1}(\mathbf{x}_i + e_{j'}, 1, \mathbf{1}_j) + v_{t-1}(\mathbf{x}_i - e_{j'}, 1, \mathbf{1}_j)) \\ (45) \quad &+ v_{t-1}(\mathbf{x}_i, 1, \mathbf{1}_j) + v_{t-1}(\mathbf{x}_i, 2, \mathbf{1}_j) \\ (46) \quad &+ j(v_{t-1}(\mathbf{x}_i, 1, \mathbf{1}_j) + v_{t-1}(\mathbf{x}_i, 2, \mathbf{1}_j)). \end{aligned}$$

By (31) from $(t-1) \rightarrow t$, $v_{t-1}(\mathbf{x}_{i-1}, 1, \mathbf{1}_j) = v_t(\mathbf{x}_{i-1}, 2, \mathbf{1}_j)$. Also (31) shows that each $\mathbf{y}_i \sim \mathbf{x}_i$ with $\mathbf{y}_i \geq 2$, $v_t(\mathbf{y}_i, 2, \mathbf{1}_j) \geq v_{t-1}(\mathbf{y}_i, 1, \mathbf{1}_j)$. If $\mathbf{x}_i - e_{j'} \not\geq 2$, $v_t(\mathbf{x}_i - e_{j'}, 2, \mathbf{1}_j) = v_t(\mathbf{x}_i, 1, \mathbf{1}_j) \geq v_{t-1}(\mathbf{x}_i, 1, \mathbf{1}_j)$. This shows that (41) $\geq$ (44). Next, (40) implies

$$(42) \geq 2v_t(x_i, 1, \mathbf{1}_j) - 2,$$

while (31) from $(t-1) \rightarrow t$ implies

$$(45) \leq 2v_{t-1}(x_i, 1, \mathbf{1}_j),$$

hence (42) $\geq$ (45). Finally, by (31) from $(t-1) \rightarrow t$,

$$v_t(x_i, 2, 2, \mathbf{1}_{j-1}) \geq v_{t-1}(x_i, 2, \mathbf{1}_j)$$

which implies (43) $\geq$ (46).

We now verify (30) for different regimes of t .

Case 1: $t < \tau_M$, $i \geq 1$

We use Lemma 2.4 as in the proof for $d = 1$ to show that

$$v_t(\mathbf{x}) - v_t(\mathbf{x} + 2e_i) \leq 2x_i$$

for all $d \leq i \leq 1$ and $\mathbf{x} \in \mathcal{S}_M$. Indeed as $t < \tau_M$

$$(47) \quad v_t(\mathbf{x}) - v_t(\mathbf{x} + e_j) \leq x_j$$

for all $\mathbf{x} \in \mathcal{S}_M$ and $1 \leq j \leq d$. Hence, $v_t(\mathbf{x}) - v_t(\mathbf{x} + 2e_j) \leq 2(M - 1)$ on $\partial_{2e_j}\mathcal{S}_M$ for all $1 \leq j \leq d$. The reflecting boundary is checked in the same way as $d = 1$, using weak facet compatibility in higher dimensions.

Case 2: $t \geq \tau_M$, $i \geq 2$

Here we show that

$$v_t(\mathbf{x}) - v_t(\mathbf{x} + 2e_i) \leq 2x_i$$

for all $d \geq i \geq 2$ and $x_1 \geq 2$. We again use Lemma 2.4 except the region in which we have the derivative bound shrinks and therefore our boundaries change. The dissipating boundary gets smaller, $\mathcal{B}^{(disp)} := \{x \in \mathcal{S}_M : x_j = M - 1 \text{ for some } 2 \leq j \leq d\}$ and the reflecting boundary remains the same except for the removal of a single point, $\mathbf{1}$. By axis monotonicity, $\sup_{x \in \mathcal{B}^{(disp)}} v_t(x) \leq v_t(M, \mathbf{1}) \leq M$. Checking the reflective boundary is as in $d = 1$ except for the point $(2, \mathbf{1}_{d-1})$. We show directly that

$$(48) \quad v_t(2, \mathbf{1}_{d-1}) \leq v_t(3, 2, \mathbf{1}_{d-2}) + 2.$$

Suppose for sake of contradiction that $v_t(2, \mathbf{1}_{d-1}) = v_t(3, 2, \mathbf{1}_{d-2}) + 2$ and $\Delta v_t(2, \mathbf{1}_{d-1}) \geq 0$ but $\Delta v_t(3, 2, \mathbf{1}_{d-1}) < 0$. As (30) has been verified for all $x \in \mathcal{S}_M$ other than $(2, \mathbf{1}_{d-1})$, (31) holds for $(3, 2, \mathbf{1}_{d-1})$ and so $v_t(3, 2, \mathbf{1}_{d-2}) = v_t(3, \mathbf{1}_{d-1})$. Then, by definition of the symmetric Laplacian, weak facet compatibility, and axis monotonicity,

$$\begin{aligned} \Delta v_t(2, \mathbf{1}_{d-1}) &= -2dv_t(2, \mathbf{1}_{d-1}) \\ &\quad + v_t(3, \mathbf{1}_{d-1}) + v_t(1, \mathbf{1}_{d-1}) \\ &\quad + (d-1)(v_t(2, \mathbf{1}_{d-1}) + v_t(2, 2, \mathbf{1}_{d-2})) \\ &\leq -2v_t(2, \mathbf{1}_{d-1}) + v_t(3, \mathbf{1}_{d-1}) + v_t(1, \mathbf{1}_{d-1}) \\ &\leq -1, \end{aligned}$$

which is a contradiction.

Weak facet compatibility: (29). The only remaining case is

$$v_t^{(d)}(1, \mathbf{1}_j) = v_t^{(d)}(2, \mathbf{1}_j) + 1.$$

By symmetry,

$$\Delta v_t^{(d)}(1, \mathbf{1}_j) = d(v_t^{(d)}(2, \mathbf{1}_j) - v_t^{(d)}(1, \mathbf{1}_j)) = -d.$$

Strong topple control: (32). We use strong topple control established in dimension $(d - 1)$. Suppose for sake of contradiction there exists $\mathbf{x} \in \mathcal{S}_M^{(d)}$ with $v_{\tau_{M-1}+2}(\mathbf{x}) - v_{\tau_{M-1}}(\mathbf{x}) = 2$. Pick $\mathbf{x} = (\mathbf{x}_{d-1}, x)$ so that $x \geq 1$ is minimal.

Case 1: $x = 1$

By dimensional reduction at time τ_{M-1} , $v_{\tau_{M-1}}^{(M,d)}(\mathbf{x}_{d-1}, 1) = v_{\tau_{M-1}}^{(M,d-1)}(\mathbf{x}_{d-1})$. By the parabolic least action principle,

$$v_{\tau_{M-1}+2}^{(M,d-1)}(\mathbf{x}_{d-1}) \geq v_{\tau_{M-1}+2}^{(M,d)}(\mathbf{x}_{d-1}, 1),$$

which contradicts (32) for $v_t^{(M,d-1)}$.

Case 2: $x = 2$

By (35),

$$v_{\tau_{M-1}}^{(M,d)}(\mathbf{x}_{d-1}, 1) = v_{\tau_{M-1}}^{(M,d)}(\mathbf{x}_{d-1}, 2),$$

which in turn, by axis monotonicity, implies $v_{\tau_{M-1}+2}^{(M,d)}(\mathbf{x}_{d-1}, 1) = v_{\tau_{M-1}}^{(M,d)}(\mathbf{x}_{d-1}, 1) + 2$, which contradicts the minimality of x .

Case 3: $x \geq 3$

Some neighbor $\mathbf{y} \sim \mathbf{x}$ must have toppled twice previously. As $x \geq 3$, $\mathbf{y} = (\mathbf{y}_{d-1}, y)$ for $y \geq 2$. The same argument for $d = 1$ when $y \geq 2$ then implies, $v_{\tau_{M-1}+1}^{(M,d)}(\mathbf{y}) = v_{\tau_{M-1}+1}^{(M-1,d)}(\mathbf{y})$ which contradicts (32) for $v_t^{(M-1,d)}$.

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AHMED BOU-RABEE, DEPARTMENT OF STATISTICS, UNIVERSITY OF CHICAGO, 5747 S. ELLIS AVENUE, CHICAGO, IL 60637, USA